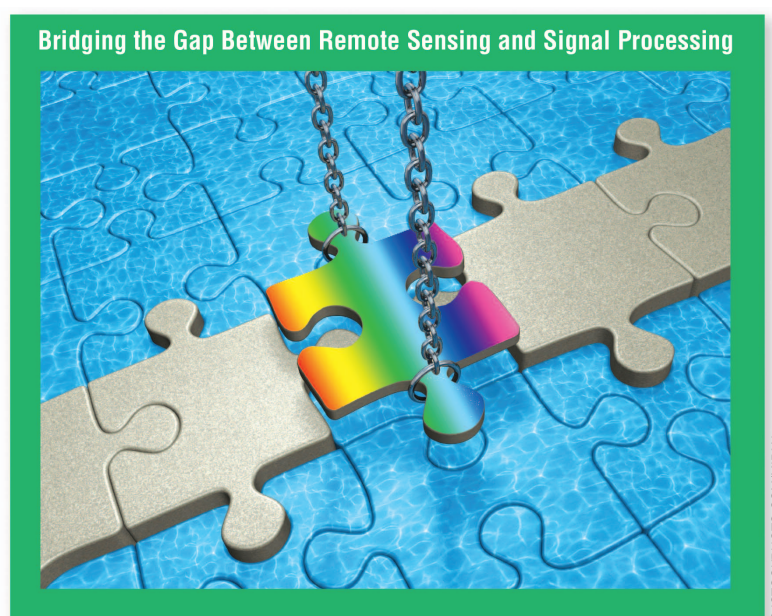


Detection Algorithms in Hyperspectral Imaging Systems



[An overview of practical algorithms]

Hyperspectral imaging applications are many and span civil, environmental, and military needs. Typical examples include the detection of specific terrain features and vegetation, mineral, or soil types for resource management; detecting and characterizing materials, surfaces, or paints; the detection of man-made materials in natural backgrounds for the purpose of search and rescue; the detection of specific plant species for the purposes of counter narcotics; and the detection of military vehicles for the purpose of defense and intelligence. The objective of this article

is to provide a tutorial overview of detection algorithms used in current hyperspectral imaging systems that operate in the reflective part of the spectrum ($0.4\text{--}2.4\ \mu\text{m}$). The same algorithms might be used in the long-wave infrared spectrum; however, the phenomenology is quite different. The covered topics and the presentation style have been chosen to illustrate the strong couplings among the underlying phenomenology, the theoretical framework for algorithm development and analysis, and the requirements of practical applications.

INTRODUCTION

In hyperspectral target detection applications, we seek to determine whether a rare object with a known spectral signature is

present or not in the imaged scene. The term *rare* is used here to signify a relatively small number compared to the total number of pixels, e.g., a few pixels in a million pixel image.

The processing methods developed for landscape classification over the last few decades are not applicable to target detection for two reasons. First, even if there are targets in the scene, their number is typically too small to support estimation of statistical properties of the target class from training data. Second, depending on the spatial resolution of the sensor, targets of interest may not be clearly resolved and hence may appear in only a few pixels or even as part of a single pixel (subpixel target). The isolated character or absence of targets from the imaged scene makes verification by means of clustering of like samples problematic. However, as we explain in this article, target detection is still possible in these situations by developing algorithms using classical detection theory and physics-based signal models. However, even the best detection algorithms exhibit a large number of false alarms; therefore, in practical systems, detected targets are further analyzed by the false alarm mitigation (FAM) and target identification (TID) algorithms. A large number of hyperspectral detection algorithms have been developed and tried over the last two decades. The reviews in [1] and [2] cover developments up to 2002; more recent developments are reviewed in [3].

The purpose of this tutorial review article is threefold. First, we introduce the physical principles that determine the nature of target and background clutter spectra measured by a hyperspectral sensor. Second, we derive the detectors currently used in practical applications, investigate their properties and performance, and discuss their practical implementation. Third, we discuss the need for FAM and TID algorithms and review some recent work in these areas. Whenever possible, we take the opportunity to mention areas where further understanding of phenomenology or better algorithms would be beneficial.

AT-SENSOR RADIANCE SIGNAL MODEL

We start with a simple discussion of the radiation components in the reflective spectral region. As shown in Figure 1, the total radiation signal reaching the sensor consists of three components:

- 1) the radiation L_1 reflected from the pixel of interest, i.e., the direct and diffused solar radiation incident on the pixel surface and reflected directly into the sensor (reflected radiance)
- 2) the radiation L_2 reflected from the surface surrounding the pixel of interest and scattered by the air volume into the sensor (adjacency radiance)
- 3) the path radiance L_3 , i.e., the photons scattered into the sensor's field of view, without ground contact.

Only the reflected radiance component L_1 contains information about the pixel of interest. The task of atmospheric correction is the calculation and removal of components L_2 and L_3 and the retrieval of the pixel reflectance from component L_1 .

Although the interactions of the radiance terms are not independent, under many atmospheric conditions and wavelengths the total spectral radiance $L(\lambda_k)$ received at the sensor can be well approximated by the linear relation [4]

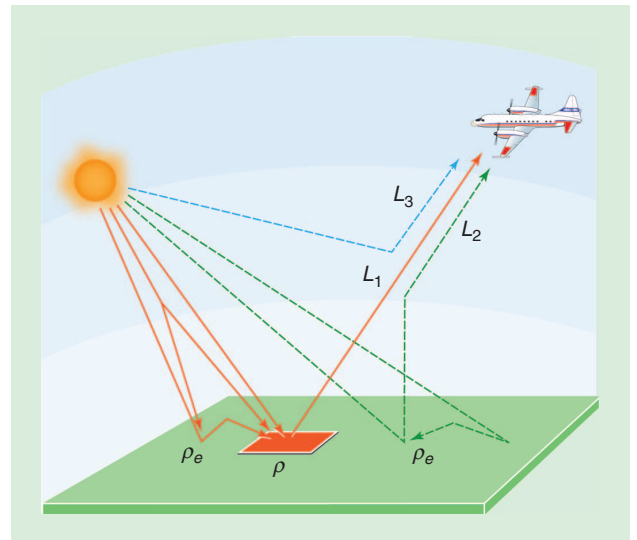
$$L(\lambda_k) = c(\lambda_k)\rho(\lambda_k) + d(\lambda_k), \quad (1)$$

where λ_k is the central wavelength of the k th spectral channel and $\rho(\lambda_k)$ is the pixel surface reflectance. The first term in (1) corresponds to L_1 ; the second term includes L_2 and L_3 .

RADIANCE AND REFLECTANCE DOMAINS

Hyperspectral imaging sensors have a sufficient number of spectral bands to allow extraction of spectra that closely resemble those acquired by laboratory spectrometers. This makes possible, at least in principle, the use of such spectra to identify materials based upon their reflectance properties and provides the basis for most hyperspectral imaging applications. Several libraries of high-resolution reflectance spectra of natural and man-made materials have been compiled by various organizations and are available for public use. We note that quite often the spectral signatures provided for various materials represent “averages” or “typical” examples. To use these library spectra, the raw sensor data must go through a series of processing steps to convert it to a usable format.

The first step in the conversion process is to calibrate the raw sensor data to the physical units of radiance. Radiance quantifies the energy at a given time passing through the input aperture of the sensor within a certain angular extent (solid angle) as a function of direction. The second step is the application of physical radiative transfer models to convert the radiance data to reflectance. This is known as atmospheric compensation. These models account for atmospheric transmission and the atmosphere's thermal emission. The difficulty with radiative transfer models lies in needing a rather detailed understanding of the atmospheric constituents. For example, the number and distribution of the small suspended particles, called aerosols, are typically unknown and often difficult to estimate. Alternatively, there are in-scene methods that assume the presence of materials with known spectral signatures [5] to enable the retrieval of the desired reflectance signature.



[FIG1] The physics-based model for at-sensor measured radiance from a Lambertian ground pixel surface.

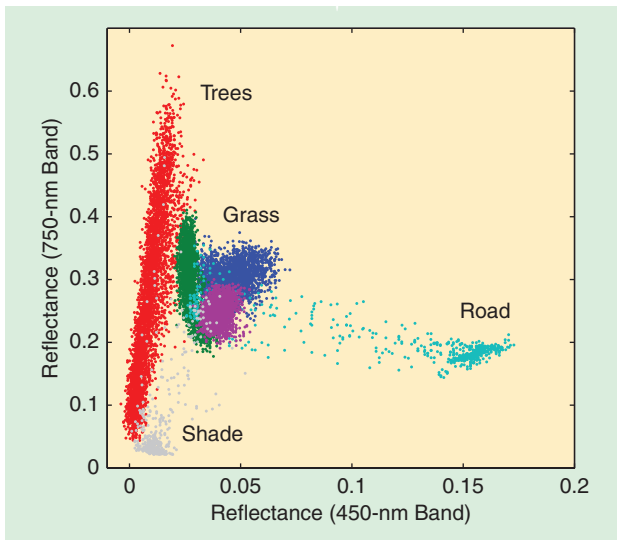


[FIG2] Areas of interest for different ground covers: trees, different types of grass, a road, and a small area dominated by shade.

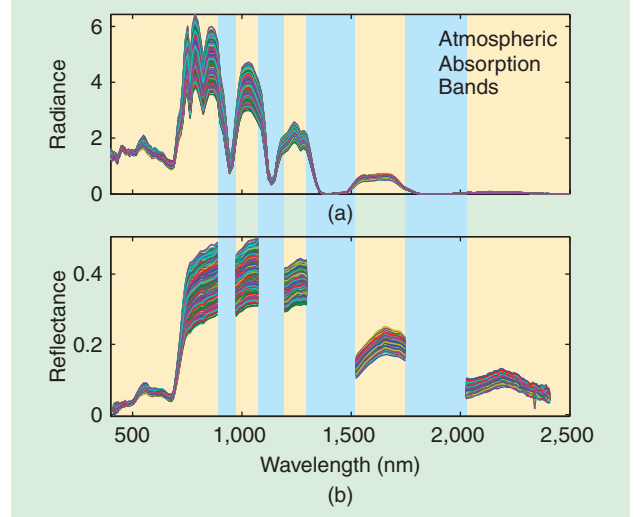
The challenge of atmospheric compensation can be understood by considering the spectra in the rectangular area enclosed by the white frame in Figure 2. Figure 3(a) shows radiance spectra for all pixels of this green area as measured by a calibrated sensor. Figure 3(b) shows the reflectance spectra after in-scene atmospheric compensation. It is obvious that the two graphs for each material appear significantly different and that the rich spectral content introduced by the atmosphere dominates the spectral radiance signatures.

SPECTRAL VARIABILITY

Figure 4 shows a scatter plot for the ground cover areas shown in Figure 2. We note that measured spectra corresponding to pixels with the same surface type exhibit a random variation



[FIG4] The scatter plot for the ground cover areas shown in Figure 2. Note that the grass class includes three different overlapping types of grass.



[FIG3] (a) and (b) Radiance and reflectance spectra retrieved from radiance extracted from the green grass area located on the left of the road in Figure 2 (white rectangular frame).

(spectral variability) that prevents the characterization of homogeneous surface materials by unique spectral signatures. The pixel-to-pixel spectral variability (background clutter) is measured by the signal-to-clutter ratio (SCR), which typically dominates sensor noise. Hence, detection algorithms primarily focus on clutter suppression.

The mathematical representation of the “spectral-swarms” in Figure 4 provides the basis for the development of hyperspectral exploitation algorithms. The statistical approach models the data clouds using probability distributions; the geometrical approach typically uses linear subspaces [2]. The geometrical approach can be justified by the linear mixing model, which states that when a pixel contains several spatially distinct materials, its spectrum is given by

$$x = a_1s_1 + \dots + a_ms_m + e = Sa + e \quad (2)$$

$$a_k \geq 0, \quad a_1 + \dots + a_m = 1, \quad (3)$$

where s_k is a “pure” spectrum, called *endmember*, a_k is the fill-fraction of each endmember, and e is the residual that accounts for the difference between the measured and modeled spectra. The mixing component Sa in (2) lies in an m -dimensional subspace of $\mathbb{R}^p$. The constraints (3), which ensure physically meaningful values for the fill-fractions, force the mixed spectra to lie on a simplex in $\mathbb{R}^p$. The enforcement of the nonnegativity constraint $a_k \geq 0$ introduces a major complication in the analysis and application of the linear mixing model.

STATISTICAL MODELING OF BACKGROUND CLUTTER

Inspection of several density scatter plots, like the one in Figure 4, suggests that a plausible model for the PDF of hyperspectral imaging data is the density mixture

$$f(x) = \sum_{k=1}^{N_c} \pi_k f_k(x), \quad \pi_k \geq 0, \quad \sum_{k=1}^{N_c} \pi_k = 1, \quad (4)$$

where π_k is the a priori probability, $f_k(x)$ is the PDF of the k th component (class), and N_C is the number of components. In classification applications, each component is modeled using a normal distribution [6]. A random vector x has a Gaussian or normal distribution, denoted by $x \sim N_p(m, C)$, if its probability density function (PDF) is given by [7]

$$f(x) = \frac{1}{(2\pi)^{\frac{p}{2}} |C|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (x - m)^T C^{-1} (x - m) \right], \quad (5)$$

where m is the mean and C is the covariance matrix.

For target detection applications, the components $f_k(x)$ must capture the heavy-tail behavior of natural hyperspectral backgrounds. A useful and flexible model that fulfills this requirement is the multivariate t -elliptically contoured distribution (t -ECD), defined by [7]

$$x = (\zeta/\nu)^{-1/2} z + m, \quad (6)$$

where $z \sim N(0, C)$ and $\zeta \sim \chi^2_\nu(0)$. When $\nu = 1$, the t -ECD is equal to the multivariate Cauchy distribution and has very heavy tails. As $\nu \rightarrow \infty$, the distribution tends toward the multivariate normal distribution and has lighter tails. Modeling of natural hyperspectral backgrounds using t -ECD mixture densities is thoroughly discussed in [8].

MODELING SUBPIXEL TARGETS

Consider a target object that occupies a fraction of the pixel area. The result is a mixed pixel whose spectrum is determined by the linear mixing model defined by (2) and (3). The mixed spectrum is given by

$$x = as + (1 - a)v, \quad 0 \leq a \leq 1, \quad (7)$$

where a is the target fill-fraction, s is the target spectrum, and v is the spectrum of the surrounding background. Equation (7) is known as a replacement target model because when the target is present, it “replaces” an equal part of the background. When $a = 1$, we have a resolved or full-pixel target, whereas if $0 < a < 1$, we have a subpixel target. For full-pixel targets, the major obstacle to detection is the variability of target and background spectra. An additional obstacle for subpixel targets is that, according to (7), the background acts as interference, which directly distorts the shape of the observed target spectrum.

Note that the additive target model $x = as + v$, used in radar, is a good approximation to the replacement target model when $a \ll 1$. The simplex for $m = 2$ is the line segment connecting the tips of s and v . For real data, the mixed pixels are clustered around this line because of spectral variability. This is clearly illustrated in Figure 4 in the case of the road and green grass classes; pixels at the edge of the road are mixed with green grass and lie between the road and grass swarms.

A statistical replacement model is obtained if we assume that the target $s \sim N(m_s, C_s)$ is mixed with one of the background classes [9]. Since the tails of the PDF have practically no effect on the probability of detection, we assume that $v \sim N(m_v, C_v)$. This results in the following distribution

$$\begin{aligned} x &\sim N(m(a), C(a)) \\ m(a) &= am_t + (1 - a)m_v \\ C(a) &= a^2 C_t + (1 - a)^2 C_v, \end{aligned} \quad (8)$$

where m_v and C_v are the parameters of the mixing class. A major practical problem with this model is that we seldom have accurate estimates of the target covariance matrix.

LIKELIHOOD RATIO DETECTORS

It is well known from statistical decision theory that decisions based upon the likelihood ratio test (LRT) are optimum over a wide range of performance criteria [10]. Among them is the Neyman–Pearson (NP) criterion, which maximizes the probability of detection for any desired probability of false alarm. The LRT is defined by

$$\Lambda(x) = \frac{f(x | \text{Target present})}{f(x | \text{Target absent})} \triangleq \frac{f(x | H_1)}{f(x | H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \eta, \quad (9)$$

where $f(x | H_i)$ is the conditional PDF under each hypothesis. Using the density mixture model (4) for $f(x | H_0)$ and the replacement model (8) for $f(x | H_1)$ leads to a computationally complicated LRT with many unknown parameters. Efforts to simplify the problem include the use of a single class for the background [9], [11] or clustering the data and using a detector for each class [12]. Despite their intuitive appeal, cluster-based detection techniques do not seem to perform better than the currently used matched filter algorithms [13].

The essential performance metric for target detection algorithm is the receiver operating characteristic (ROC) curve. Reliable estimation of P_D requires between 50 and 100 target pixels with approximately the same fill-fraction. This costly requirement has limited the availability of data sets for the assessment of target detection algorithms. Therefore, we often resort to performance prediction models or we pick a threshold that allows detection of all targets and we count the number of resulting false alarms.

For a fully fair comparison of detection algorithms, we should take the position of an operational user and consider how the use of each detector in a practical application affects its performance and utility. Two very important practical requirements are good performance on a diverse variety of targets and backgrounds and operational ease of use by nonexperts. Furthermore, computational complexity, numerical robustness, robustness to the selection of free parameters, and operation without a man-in-the-loop are also critical for surveillance applications.

MATCHED FILTER DETECTION THEORY

The matched filter algorithms used in current systems are often derived in the hyperspectral literature using the additive model [12], which is a good approximation for small fill-fractions and a bad one for full-pixel targets. In [1], we partially bypass this inconsistency by providing separate derivations for full- and subpixel targets. In this article, we derive the matched filter detectors without any attempt to choose a “good” signal model for the data. Instead, we use the shape of the decision surface to choose algorithms and explain their suitability for hyperspectral target detection.

Consider a $p \times 1$ observation vector x with distribution

$$x \sim N_p(as, \sigma_b^2 C_b), \quad (10)$$

where $a \geq 0$ and $\sigma_b^2 \geq 0$ are scalar quantities, s is a $p \times 1$ vector, and C_b is a $p \times p$ positive definite matrix.

The objective is to find an NP test (detector) to choose between the following hypotheses:

$$H_0: a = 0 \quad (\text{target absent}) \quad (11a)$$

$$H_1: a > 0 \quad (\text{target present}). \quad (11b)$$

The vector s and the matrix C_b are always assumed known. Since the matrix C_b is positive definite, its square-root decomposition $C_b = C_b^{1/2} C_b^{1/2}$ is invertible. Therefore, the whitening transformation

$$\tilde{x} = C_b^{-1/2} x, \quad \tilde{s} = C_b^{-1/2} s \quad (12)$$

results in the random vector

$$\tilde{x} \sim N_p(a\tilde{s}, \sigma_b^2 I) \quad (13)$$

with spherical (“white”) normal distribution. The detection problems specified by (10) and (1) or by (13) and (1) are equivalent; thus, we use the spherical model (13) to simplify subsequent derivation and analysis.

If the distribution in (13) is fully specified, i.e., we know a and σ_b^2 , the likelihood ratio is given by $\Lambda(\tilde{x}) = f(\tilde{x}|H_1)/f(\tilde{x}|H_0)$. If $\Lambda(\tilde{x})$ exceeds a certain threshold η we accept H_1 as true; otherwise, we accept H_0 as true. Since any monotonic function of $\Lambda(\tilde{x})$ results in equivalent decisions, the LRT for the signal model (13) leads to the following test:

$$y_1 = a\tilde{s}^T \tilde{x} \underset{H_0}{\overset{H_1}{\gtrless}} \eta_1 \sigma_b^2. \quad (14)$$

If $a > 0$ under H_1 , we can divide both sides of (14) by a without affecting the direction of the inequality. The resulting statistic

$y_2 = \tilde{s}^T \tilde{x}$ is completely independent of a . In conclusion, if a is unknown but $a > 0$, the matched filter (MF) detector, which is usually defined by the normalized expression

$$y_{\text{MF}} = \frac{\tilde{s}^T \tilde{x}}{\sqrt{\tilde{s}^T \tilde{s}}} = \frac{s^T C_b^{-1} x}{\sqrt{s^T C_b^{-1} s}} \underset{H_0}{\overset{H_1}{\gtrless}} \eta_{\text{MF}} \quad (15)$$

is NP-optimum for all values $a > 0$. Therefore, the MF detector (15) is a uniformly most powerful (UMP) test. The MF does not have the constant false alarm rate (CFAR) property because the threshold depends on both a and σ_b^2 .

If a and σ_b^2 are unknown, an intuitively appealing approach is to use their maximum likelihood estimates (MLEs) to form a generalized LRT (GLRT). The MLEs of a and σ_b^2 in (13) are

$$\hat{a} = \frac{\tilde{s}^T \tilde{x}}{\tilde{s}^T \tilde{s}}, \quad p\hat{\sigma}_b^2 = \tilde{x}^T \tilde{x} - \frac{(\tilde{s}^T \tilde{x})^2}{\tilde{s}^T \tilde{s}}. \quad (16)$$

The resulting maximum value of the likelihood function is $\exp(-p/2)/(2\pi\hat{\sigma}_b^2)^{p/2}$. Therefore, the GLR is

$$\Lambda_G(\tilde{x}) = \left(\frac{\hat{\sigma}_b^2 \text{ under } H_1}{\hat{\sigma}_b^2 \text{ under } H_0} \right)^{-\frac{p}{2}} = \left[1 - \frac{(\tilde{s}^T \tilde{x})^2}{(\tilde{s}^T \tilde{s})(\tilde{x}^T \tilde{x})} \right]^{-\frac{p}{2}}. \quad (17)$$

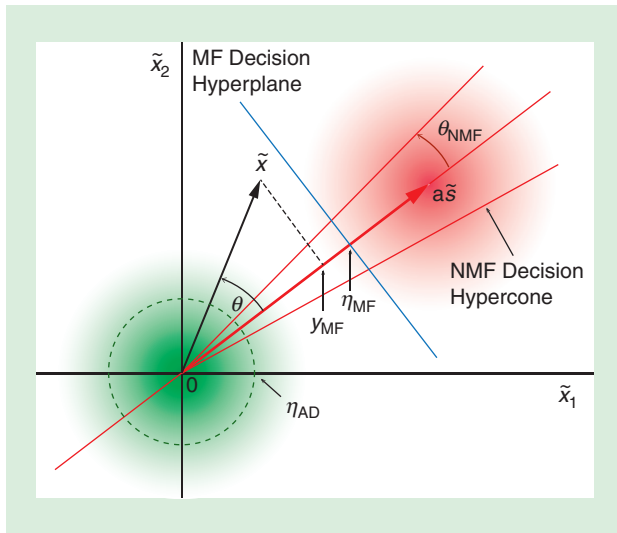
Since $a > 0$ under H_1 and only the term $\tilde{s}^T \tilde{x}$ retains the sign of $\hat{a}$, the normalized MF (NMF), defined by

$$y_{\text{NMF}} = \frac{\tilde{s}^T \tilde{x}}{\sqrt{\tilde{s}^T \tilde{s}} \sqrt{\tilde{x}^T \tilde{x}}} = \frac{s^T C_b^{-1} x}{\sqrt{s^T C_b^{-1} s} \sqrt{x^T C_b^{-1} x}} \underset{H_0}{\overset{H_1}{\gtrless}} \eta_{\text{NMF}} \quad (18)$$

is a logical test for the hypotheses (11). The NMF is also known as a CFAR matched filter [14]. Figure 5 provides an intuitive geometrical interpretation of MF and NMF detectors in the whitened space. The vector $\tilde{s}/\|\tilde{s}\|$, where $\|\cdot\|$ denotes Euclidean length, is a unit vector; hence, the value of y_{MF} is the scalar projection of the observation vector onto the target. In contrast, the value of y_{NMF} is the cosine of the angle between the observation and target vectors. Therefore, the NMF is invariant to scaling of the target and observation measurements and has the CFAR property. In fact, the NMF has the CFAR property for all elliptically contoured distributions. The NMF is uniformly most powerful only for the class of detectors that are invariant to rotations about $\tilde{s}$ and to scaling of target and observation spectra [10]; therefore, under the signal model (13), it cannot perform better than the UMP MF detector. This must be expected because the NMF assumes that σ_b^2 is unknown, i.e., the NMF uses “less” information.

To understand the operation of MF and NMF detectors in the whitened space, we look at their decision boundaries, which are obtained by equating (15) and (18) to a constant threshold value. The decision surface for the MF is a hyperplane perpendicular to $\tilde{s}$ at a location determined by the threshold η_{MF} . Note that all observations vectors with tips “touching” this hyperplane have the same MF response. The decision surface for the NMF is the surface of a cone with vertex at the origin and axis along the vector $\tilde{s}$. The vertex angle θ_{NMF} is specified by the threshold η_{NMF} .

If the mean value of x in (10) lies in a known subspace, i.e., $E(x) = S_t a_t$, where S_t is a $p \times q_t$ matrix, the GLRT for $a_t = 0$ versus $a_t \neq 0$ is given by [1]



[FIG5] Geometrical interpretation of MF, NMF, and AD decision surfaces in the whitened-observation space. These decision surfaces determine the performance of each detector for any situation.

$$T(x) = x^T C_b^{-1} S_t (S_t^T C_b^{-1} S_t)^{-1} S_t^T C_b^{-1} x. \quad (19)$$

If $q_t = p$ and S_t has full rank, the target subspace spans the entire observation space, and the GLRT statistic (19) is reduced to the square-law detector

$$y_{AD} = x^T C_b^{-1} x = \tilde{x}^T \tilde{x} \stackrel{H_1}{\underset{H_0}{\gtrless}} \eta_{AD}, \quad (20)$$

which is also known as the RX anomaly detector (AD) [15]. We note that the MF and NMF detectors use the target signature s to determine a “preferred” search-direction in the whitened-observation space. Since the AD does not use the target signature, there is no preferred direction; the decision is based solely on the Mahalanobis distance $x^T C_b^{-1} x$ or the Euclidean distance $\|\tilde{x}\|$. Thus, the decision surface of AD is a hypersphere in the whitened-observation space (see Figure 5). Overviews of AD algorithms are provided in [3], [16], and [17].

IDEAL PERFORMANCE ANALYSIS

Detection performance is commonly evaluated in terms of ROC curves. The use of the scaled covariance $\sigma_b^2 C_b$ in the signal model (10) was necessary for the derivation of the NMF detector. To simplify subsequent derivations and notation, without any loss of generality, we assume $\sigma_b^2 = 1$ for the rest of the article.

We first determine the PDFs under the true model (13), which ensures the best theoretical performance for each detector. Since spherical distributions are invariant under orthogonal transformations (rotations) [7], to simplify the derivations, we use the transformation

$$z = Q^T \tilde{x}, \quad Q^T = Q^{-1}. \quad (21)$$

The orthogonal matrix Q is chosen such that

$$Q^T \tilde{s} = \|\tilde{s}\| e_1 \quad (22)$$

with $e_1 = [1 \ 0 \dots 0]^T$. The first column of Q is the unit vector $q_1 = \tilde{s}/\|\tilde{s}\|$; the remaining columns are arbitrarily chosen to make Q an orthogonal matrix. Since

$$z_1 = q_1^T \tilde{x} = \frac{\tilde{s}^T \tilde{x}}{\|\tilde{s}\|} = y_{MF}, \quad (23)$$

the effect of this rotation is to fit the response of the MF along the direction of the target vector in the whitened-observation space and to fit the remaining coordinates of z along the remaining “target-free” directions.

The distribution of the rotated observation vector is

$$z \sim N_p(a\|\tilde{s}\|e_1, I). \quad (24)$$

Using (20), (23), and (24) and the definition of the chi-squared distribution, we obtain

$$y_{MF} = z_1 \sim N(a\|\tilde{s}\|, 1) \quad (25)$$

$$y_{AD} = \|\tilde{x}\|^2 = \|Qz\|^2 = \|z\|^2 \sim \chi_p^2(a^2\|\tilde{s}\|^2). \quad (26)$$

To derive the distribution of the NMF, we note that

$$y_{NMF} = \frac{z_1}{\|z\|^2} = \frac{t}{\sqrt{p-1+t^2}}, \quad t \triangleq \sqrt{p-1} \frac{z_1}{\|z_2\|}, \quad (27)$$

where $z_2 \triangleq [z_2 \dots z_p]^T$. From (27) and the definition of t -distribution, we have

$$t = \frac{z_1}{\frac{1}{\sqrt{p-1}}\|z_2\|} \sim t_{p-1}(a\|\tilde{s}\|), \quad (28)$$

which is a noncentral t -distribution. The PDFs under H_1 (target present) hypothesis depend on the SCR

$$\text{SCR} \triangleq a^2\|\tilde{s}\|^2 = a^2 s^T C_b^{-1} s, \quad (29)$$

which is the Mahalanobis distance between the means under the two hypotheses. Note that $a = 0$ under the H_0 hypothesis. Thus, we can use these PDFs to evaluate the performance of MF, NMF, and AD using ROC curves.

PERFORMANCE UNDER MODEL MISMATCHES

Application of the MF and NMF assumes perfect knowledge of the in-scene target signature s and the clutter covariance matrix C_b . We now analyze performance losses when the signature s_0 and covariance C_0 used by the detectors are

$$s_0 \neq s \quad (\text{Signature mismatch}) \quad (30)$$

$$C_0 = C_b + a^2 s s^T \quad (\text{Target in covariance}). \quad (31)$$

We first recall that the linear filter $y = h^T x$ that maximizes the output SCR, defined by

$$\text{SCR}_{\text{out}} = \frac{[E(y|H_1) - E(y|H_0)]^2}{\text{Var}(y|H_0)} = \frac{a^2 (h^T s)^2}{h^T C_b h}, \quad (32)$$

is the MF $h = \kappa C_b^{-1} s$, where κ is a normalization constant; the resulting maximum output SCR is given by (29).

Equivalently, the MF minimizes the output variance $\text{Var}(y) = \text{Var}(h^T x) = h^T C_b h$ while keeping the gain in the direction of the target constant. Thus, the MF solves the optimization problem [18]

$$\min_h h^T C_b h \quad \text{subject to} \quad h^T s = 1 \quad (33)$$

with the inconsequential change that $\kappa = 1/s^T C_b^{-1} s$.

If we use the MF $h_1 = C_b^{-1} s_0 / s_0^T C_b^{-1} s_0$, the output SCR is given by [19]

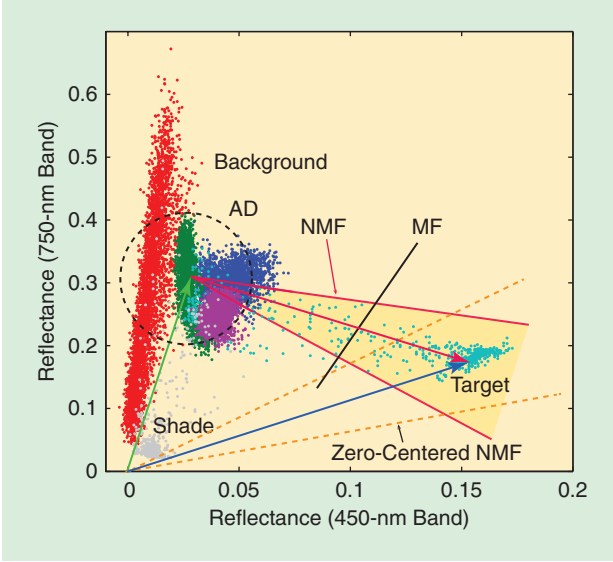
$$\text{SCR}_1 = \cos^2(s, s_0; C_b^{-1}) \cdot \text{SCR}, \quad (34)$$

where $\cos^2(s, s_0; C_b^{-1})$ is the generalized cosine defined by

$$\cos^2(s, s_0; C_b^{-1}) \triangleq \frac{(s^T C_b^{-1} s_0)^2}{(s^T C_b^{-1} s)(s_0^T C_b^{-1} s_0)} \quad (35)$$

and $-1 \leq \cos^2(s, s_0; C_b^{-1}) \leq 1$. The performance loss depends on the mismatch factor $\cos^2(s, s_0; C_b^{-1})$, but it is independent of the strength a of the target. If the target is present in the clutter covariance matrix, [see (31)], the MF is defined by $h_2 = C_0^{-1} s_0 / s_0^T C_0^{-1} s_0$. The output SCR is given by [19]

$$\text{SCR}_2 = \text{SCR}_1 [1 + (2\text{SCR} + \text{SCR}^2) \sin^2(s, s_0; C_b^{-1})]^{-1}, \quad (36)$$



[FIG6] The decision cone (yellow area) of the NMF includes the plurality of subpixel and full-pixel “road” targets (cyan dots). Note the importance of “centering” the cone at the mean of the clutter cloud.

where $\sin^2(s, s_0; C_b^{-1}) = 1 - \cos^2(s, s_0; C_b^{-1})$. The loss due to covariance contamination (target-in-covariance) is determined by the term $(2\text{SCR} + \text{SCR}^2) \sin^2(s, s_0; C_b^{-1})$, which shows a strong dependence on the target strength through SCR and on mismatch through $\sin^2(s, s_0; C_b^{-1})$. Thus for strong targets, the losses can be significant.

Similar conclusions can be drawn by performance evaluation of both the MF and NMF detectors using ROC curves under the Gaussian model (10). Similar analysis also shows that the NMF, which uses angular information, is more sensitive to mismatch than the MF [20].

Although the effects of target mismatch and covariance contamination have been extensively analyzed, there are not many practical techniques for their mitigation. A plausible way to reduce mismatch losses is to relax the “line-pointing” constraint in (33). This can be done by changing the constraint in (33) to form the following optimization problem:

$$\min_h h^T C_b h \quad \text{subject to} \quad h^T s \geq 1 \quad (37)$$

for all signatures s in the hypersphere $\|s - s_0\|^2 \leq \epsilon^2$. The value of ϵ determines the “selectivity” of the detector. Several solutions to (37) have been proposed in the literature [18]; the most practical ones lead to some kind of diagonal loading $C_b + \xi^2 I$. An interesting solution in [21] provides a direct link between ϵ and the diagonal loading factor ξ . The resulting MFs are robust to signal mismatch and covariance contamination in exchange for a higher false alarm rate.

PRACTICAL HYPERSPECTRAL DETECTORS

The scatter plot in Figure 4 and the decision surfaces in Figure 5 provide insight for the selection of hyperspectral target detection algorithms. We note that an NMF with a vertex at the center of the

grass cloud and axis pointing to the road cloud is the natural choice for the detection of subpixel and full pixel “road” targets. This is illustrated in Figure 6, which also clearly shows why a zero-centered NMF would not be a good choice for subpixel targets. However, we must stress that the ultimate criterion of success is performance in real applications.

The clutter-centered MF and NMF are obtained by subtracting the clutter mean from (15), (18), and (20). This yields

$$y_{\text{MF}} = \frac{(s - m_b)^T C_b^{-1} (x - m_b)}{\sqrt{(s - m_b)^T C_b^{-1} (s - m_b)}} \quad (38)$$

$$y_{\text{AD}} = (x - m_b)^T C_b^{-1} (x - m_b) \quad (39)$$

$$y_{\text{NMF}} = \frac{y_{\text{MF}}}{\sqrt{y_{\text{AD}}}}. \quad (40)$$

In practice, the mean m_b and covariance C_b of the background clutter are replaced by their maximum likelihood estimates

$$\hat{m}_b = \frac{1}{N} \sum_{k=1}^N x_k, \quad \hat{C}_b = \frac{1}{N} \sum_{k=1}^N (x_k - \hat{m}_b)(x_k - \hat{m}_b)^T. \quad (41)$$

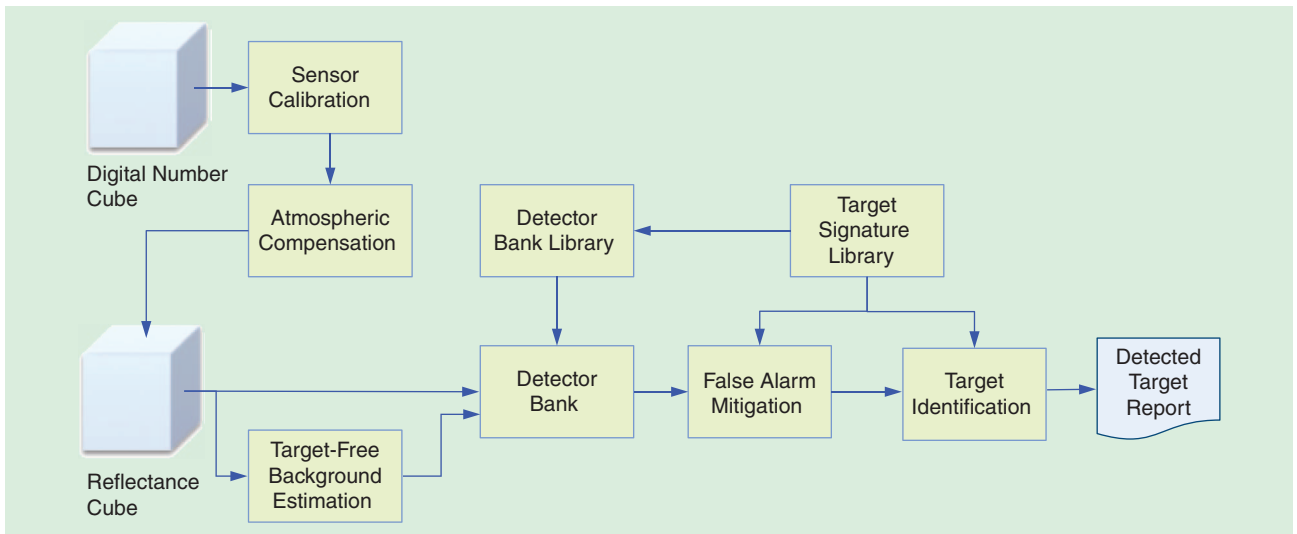
The square of (38) is known as the adaptive MF (AMF) [22]. The square of (40) is known as the adaptive NMF (ANMF) [23] or adaptive cosine/coherence estimator (ACE) [24]. The term *adaptive* is often used to emphasize that the clutter moments are estimated from the data.

We shall use a simple example to illustrate some challenges related to the application and performance evaluation of AMF and ANMF detectors using a data set from the Hyperspectral Digital Imagery Collection Experiment (HYDICE) Forest Radiance I experiment [2]. Figure 7 shows the basic components of a typical target detection system. The cube of raw digital numbers is calibrated to a radiance cube, which is then converted to reflectance to allow the use of target signatures from a library. We focus on a green tarp target having the library signature shown in Figure 8; all other targets have been removed from the data. There are three different sizes of green tarps in the scene; the smallest tarp corresponds to a subpixel target.

We first note from Figure 8 that there is a mismatch between library and in-scene target spectra due mainly to material variability, atmospheric compensation errors, sensor calibration limitations, and mixing for subpixel targets. Furthermore, there is mismatch between the normal distribution assumed in (10) and the distribution of the multiclass background of Forest Radiance I scene. Since the presence of targets in the estimate $\hat{C}_b$ decreases performance, we use the AD algorithm (20) to remove “targetlike” pixels from the training data (target-free background estimation).

In practice, $\hat{C}_b$ is often ill conditioned, and its small eigenvalues and corresponding eigenvectors are difficult to estimate and hard to compute accurately even if $N \gg p$. Numerically stable and more robust detectors can be formed by eliminating the smaller eigenvalues through averaging and loading. We often use dominant mode rejection combined with diagonal loading [25], [26].

Figure 9 shows images of the scene and the responses of the AD, MF, and NMF detectors for each pixel. Inspection of these detection statistics indicates that the MF and NMF score high at



[FIG7] A block diagram of a typical hyperspectral target detection system.

the target pixels; however, we stress that visual assessment is not a dependable way to evaluate detection performance.

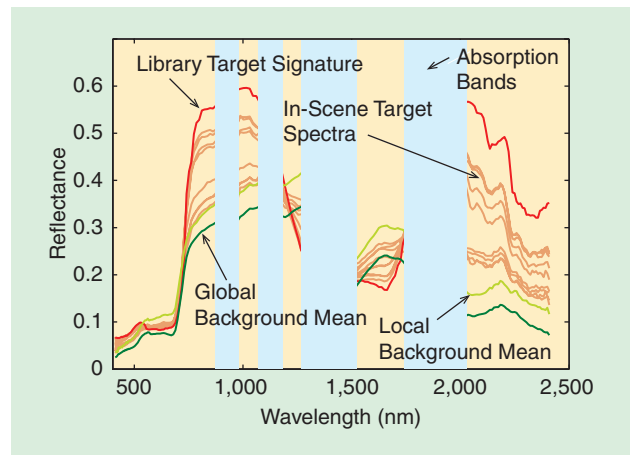
PERFORMANCE FOR MULTICLASS BACKGROUNDS

If we model the background clutter with a density mixture of t -ECDs and the target with the statistical replacement model, we can develop a model to investigate the performance of MF and NMF detectors under more realistic scenarios [27]. Needless to say, in practice, it is impossible to find PDFs that capture all aspects of hyperspectral imaging data. The presence of man-made structures in a scene can produce false alarms that cannot be predicted by any model.

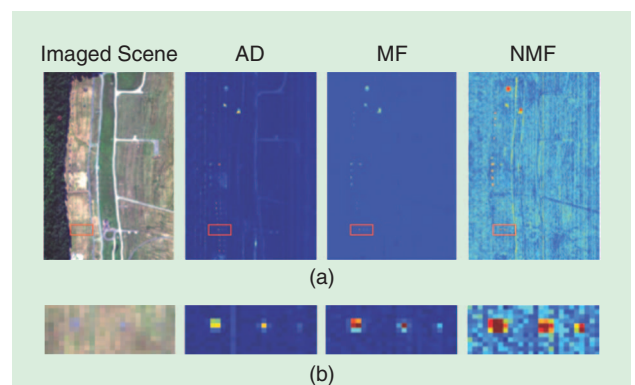
The multiclass model has been used to develop a performance prediction tool useful in obtaining insight and evaluating sensors and detection algorithms under different deployment conditions. Another interesting and useful application of the model is shown in Figure 10, which shows a scatter plot of NMF versus MF responses for the Forest Radiance I data set. The ribbon shows a 95% confidence interval for the NMF and MF responses predicted by a five-class t -ECD model. We first note that the model does a reasonable job at predicting the range of detector responses and the majority of background responses. Clearly, the NMF provides a better separation between target and clutter pixels compared to the MF. Most pixels that exceed the MF threshold belong to other targets and man-made objects present in the scene. We again stress that such results do not prove the validity of the model; a thorough validation of the predictive model, which is work in progress, is a laborious and expensive undertaking.

FALSE ALARM MITIGATION ALGORITHMS

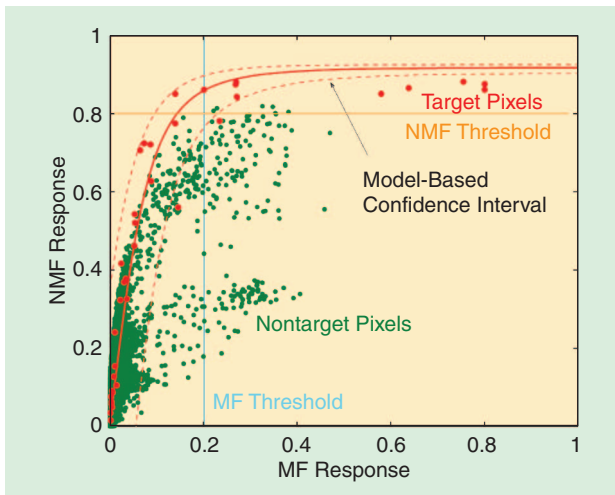
In practice, all detection algorithms exhibit a large number of false alarms [28]. A promising approach to deal with this challenge is postprocessing of the top hits with FAM algorithms [11]. The NMF and MF detectors make decisions by exploiting the statistical distributions of targets and clutter in the clutter-whitened space. Practical experience has shown that both algorithms are relatively



[FIG8] Reflectance spectra for the library target signature, the in-scene target spectra, the local background mean around the targets, and the global background mean for the Forest Radiance I data cube.



[FIG9] Imaged scene and detection statistics (detector response) for the AD, MF, and NMF algorithms. The images in (b) show the area around the targets of interest, which correspond to (a) the red frame box in the large images.



robust to model mismatches, including clutter multimodality, covariance estimation, and target signature mismatch. Therefore, FAM algorithms must exploit information not available in the detection statistic. In this sense, the development of FAM algorithms is an art involving statistics, spectroscopy, and spatial image processing. False alarms can also be mitigated using information from coregistered high-resolution video.

$$\Delta^2(x) = \|\tilde{x} - \hat{a}\tilde{s}\|^2 = (x - \hat{a}s)^T C^{-1}(a) (x - \hat{a}s) \quad (42)$$

is inversely proportional to the likelihood that the test pixel is a subpixel target. The approximation $C(a) \simeq C_b$ works reasonably well in practical applications. Figure 12 shows a typical scatter plot of MD versus MF response for a detection experiment involving a man-made target. We note that with the use of a second threshold, we can significantly reduce the number of false alarms. The use of a CFAR threshold is possible by modeling the distribution of (42). More details about the MF with FAM can be found in [11].

TARGET IDENTIFICATION

The task of a TID algorithm is to determine whether a pixel picked out by the detector contains indeed the target of interest. This is a difficult undertaking because different scene components may result to similar or identical spectra (mimics). The problem is more prominent for subpixel targets because of spectral mixing. In this case, the first step is to estimate the target spectrum using unmixing techniques. Identification algorithms use the estimated spectrum and material spectral libraries to associate this spectrum with the reference spectrum of some material. Typical approaches exploit ideas from the areas of linear unmixing, F-test-based model selection, Bayesian model selection [29], [30], band selection techniques [31], and spectroscopy [32].

CONCLUSIONS

The development of practical hyperspectral target detection algorithms requires the understanding and joint exploitation of phenomenology, sensor technology, and statistical signal processing. Target variability, the mismatch between library and in-scene signatures, and false alarms from spectral “mimics” constitute major challenges to be overcome by future algorithms. Therefore, the development of FAM and TID algorithms is expected to be an active area of research in hyperspectral target detection.

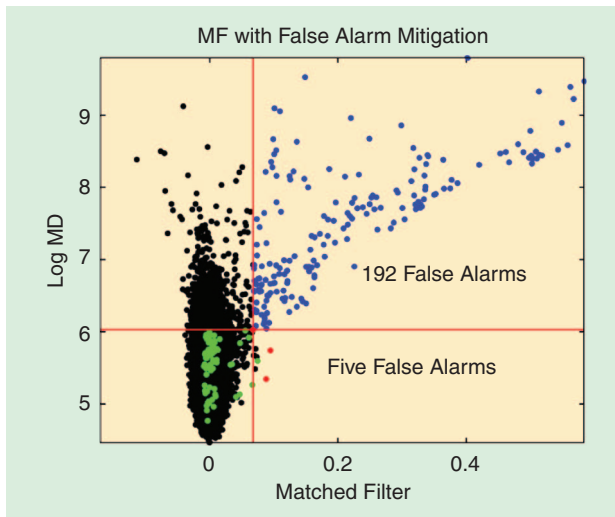
ACKNOWLEDGMENTS

This work is sponsored by the Department of the Air Force under Air Force Contract FA8721-05-C-0002. Opinions, interpretations, conclusions, and recommendations are those of the author and are not necessarily endorsed by the United States Government.

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[FIG12] The MF with FAM uses a “subpixel likelihood metric” and double thresholding to reduce false alarms.

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